Siddiqui et al. 2021 — 6-Species TMCMC Re-calibration

10,000-particle GPU-accelerated Bayesian inference (27 parameters)

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1 Dataset

Siddiqui DA, Fidai AB, Natarajan SG, Rodrigues DC (2022). "Succession of Oral Bacterial Colonizers on Dental Implant Materials: An In Vitro Biofilm Model." *Dental Materials* 38(2):384–396. DOI: 10.1016/j.dental.2021.12.021. PMC8828709.

6 species: *S. oralis*, *A. naeslundii*, *A. actinomycetemcomitans*, *V. parvula*, *F. nucleatum*, *P. gingivalis*. Planktonic + adherent on polished cpTi, 6 timepoints over 21 days. No antibiotics. Pure in vitro colonization succession.

1.1 Species and Socransky Classification

Table 1: Bacterial species in Siddiqui et al. model

Species	Abbrev	Socransky complex	Role
Streptococcus oralis	So	Yellow	Early colonizer
Actinomyces naeslundii	An	Blue	Early colonizer
Aggregatibacter actinomycetemcomitans	Aa	Green	Early colonizer
Veillonella parvula	Vp	Purple	Secondary (metabolic bridge)
Fusobacterium nucleatum	Fn	Orange	Secondary (physical bridge)
Porphyromonas gingivalis	Pg	Red	Late colonizer (pathogen)

1.2 Experimental Design

Siddiqui et al. prepared commercially pure titanium (cpTi) and yttria-stabilized zirconia (ZrO_2) disks with three surface treatments—polished, acid-etched, and sandblasted—yielding six substrate conditions in total. In the present re-calibration study, we focus on the **planktonic fraction** and the **adherent biofilm on polished cpTi**, as these represent the two extremes of bacterial habitat (bulk medium vs. surface-attached community).

The bacteria were cultured anaerobically (10% CO_2, 5% H_2, 85% N_2) at 37°C in Brain Heart Infusion (BHI) broth supplemented with hemin (5 mg/mL) and vitamin K1 (0.1 mg/mL). Each species was inoculated at 1.67×10^6 CFU/mL (total: 10^7 CFU/mL). Culture medium was replaced daily by withdrawing 2.5 mL and adding 2.7 mL of fresh media, simulating nutrient replenishment analogous to gingival crevicular fluid (GCF) flow.

Bacterial composition was quantified by species-specific quantitative PCR (qPCR) targeting single-copy genes: GDH1 (*S. oralis*), rpoB (*A. naeslundii*, *V. parvula*), waaA (*A. actinomycetemcomitans*, *P. gingivalis*), and GGT (*F. nucleatum*). Amplification efficiencies ranged from 91.1% to 100.7% with off-target amplification below 0.1%, ensuring reliable species-level resolution. Sampling was performed at six timepoints: 6 h, 1, 3, 7, 14, and 21 days post-inoculation.

1.3 Published Succession Pattern

The succession dynamics reported by Siddiqui et al. are summarized in Tables 2 and 3. The planktonic fraction showed a classic early-to-late colonizer transition: *S. oralis* dominated at 6 h (∼75%), was overtaken by the bridging species *F. nucleatum* by Day 3 (∼74%), and was ultimately supplanted by the red-complex pathogen *P. gingivalis* at Day 21 (∼60%). The transition from Day 7 to Day 14 was described as "drastic," with *P. gingivalis* increasing 100-fold as it exploited the ecological niche prepared by *F. nucleatum*—consistent with the Hill-gated Fn→Pg mechanism in our Hamilton ODE model.

The adherent fraction on polished cpTi exhibited a slower and more complex dynamics. A notable observation was the stagnation of *S. oralis* at Day 3 on polished cpTi (statistically significant vs. sandblasted ZrO_2), followed by a 10-fold recovery at Day 7. By Day 21 the adherent composition reached a balanced state with approximately equal representation of early colonizers (∼33%), secondary colonizers (∼33%), and *P. gingivalis* (∼33%). This non-monotonic behavior presents a particular challenge for ODE-based modeling, as the Day 7 reversal requires interaction dynamics beyond simple competitive exclusion.

Table 2: Planktonic succession (digitized from Siddiqui et al. Figs. 1–2 and text)

Day	So	An	Aa	Vp	Fn	Pg
0.25	0.75	0.01	0.23	0.00	0.01	0.00
1	0.42	0.03	0.05	0.12	0.38	0.00
3	0.08	0.03	0.02	0.12	0.74	0.01
7	0.12	0.05	0.03	0.12	0.65	0.03
14	0.08	0.03	0.02	0.10	0.44	0.33
21	0.03	0.05	0.02	0.05	0.25	0.60

Table 3: Adherent succession on polished cpTi (digitized)

Day	So	An	Aa	Vp	Fn	Pg
0.25	0.88	0.01	0.09	0.00	0.02	0.00
1	0.58	0.04	0.08	0.08	0.22	0.00
3	0.28	0.04	0.08	0.15	0.45	0.00
7	0.58	0.04	0.08	0.10	0.20	0.00
14	0.48	0.04	0.08	0.10	0.15	0.15
21	0.22	0.05	0.06	0.10	0.24	0.33

1.4 Comparison with Heine 2025

The Siddiqui 2021 dataset differs from our primary calibration data (Heine et al. 2025) in several important respects (Table 4). Most critically, Siddiqui uses six species including *A. actinomycetemcomitans* (green complex), which is absent in the Heine 5-species model. There is no commensal/dysbiotic manipulation—only a single culture condition representing natural succession. The substrate (polished cpTi) and *Veillonella* species (*V. parvula* vs. *V. dispar*) also differ, though these are considered interchangeable in oral biofilm models. Both studies share the absence of antibiotics and a 21-day observation window with comparable timepoint spacing.

Table 4: Experimental design comparison

	Heine 2025	Siddiqui 2021
Species	5 (So, An, Vd, Fn, Pg)	6 (+Aa)
Conditions	4 (CS/CH/DS/DH)	1 (single culture)
Manipulation	Nutrient media	None (pure succession)
Substrate	Hydroxyapatite	Polished cpTi
Veillonella	*V. dispar*	*V. parvula*
Inoculum	Mixed (equal)	Mixed (1.67×10^6/sp)
Quantification	16S rRNA (FISH)	qPCR (species-specific)
Antibiotics	No	No

1.5 Limitations of the Source Data

The authors noted several limitations that also affect our re-calibration. The simplified 6-species model excludes *Corynebacterium* spp., recently identified as a key architect of oral biofilm spatial structure in vivo. Standard qPCR amplifies genomic DNA from both live and dead cells, potentially overestimating viable counts. The BHI-based culture system yields bacterial densities of $10^{8–9}$ CFU/mL, two to three orders of magnitude higher than typical oral cavity counts ($10^{5–6}$ CFU/mL), which may compress inter-species competitive dynamics. Furthermore, the model lacks salivary pellicle formation, host cell interactions, and oxygen gradients characteristic of the supragingival environment. These factors should be considered when interpreting the inferred interaction parameters.

Replication: The study used "one sample of each type of surface-treated cpTi or ZrO_2 disk" per well, with qPCR run in "technical duplicates." No explicit biological replicates (independent culture experiments) are reported. The error bars in Figures 1–2 therefore likely reflect technical qPCR variance ($n = 2$), not biological variability. This means our digitized mean values represent a **single experimental realization**, and the true uncertainty in species composition is larger than the qPCR error bars suggest. We account for this by using a relatively wide observation noise ($\sigma_{\text{obs}} = 0.15$) in the TMCMC likelihood.

1.6 Observable: $\bar{\varphi}_i = \varphi_i \cdot \psi_i$

The Hamilton ODE evolves two sets of internal variables per species: the volume fraction φ_i and the fitness variable ψ_i. Neither is directly measurable. The experimentally observable quantity is the species abundance fraction, which corresponds to the product $\bar{\varphi}_i = \varphi_i \cdot \psi_i$ (normalized to sum to unity). This is what qPCR measures: the relative DNA copy number proportional to each species' biomass.

Separating φ_i and ψ_i from data would require independent measurements—e.g., FISH/confocal microscopy for spatial volume fractions and metabolic activity assays (CTC staining, RNA-based methods) for fitness. Neither was performed in the Siddiqui study, nor in Heine 2025. Consequently, the interaction matrix A_{ij} and growth rates μ_i are inferred through the composite observable $\bar{\varphi}_i$ only. The Hamilton variational structure constrains the φ–ψ relationship, preventing complete non-identifiability, but some parameter trade-offs between A_{ij} and μ_i are expected in the posterior.

2 Results

Table 5: Fit quality (TMCMC with vmap-GPU RW mutation)

Dataset	RMSE	Cosine	Spearman ρ	max $\log L$
Planktonic	0.0856	0.9623	0.5767	-6.7
Adherent	0.0965	0.9283	0.6932	-13.9

3 6-Panel Posterior Predictive

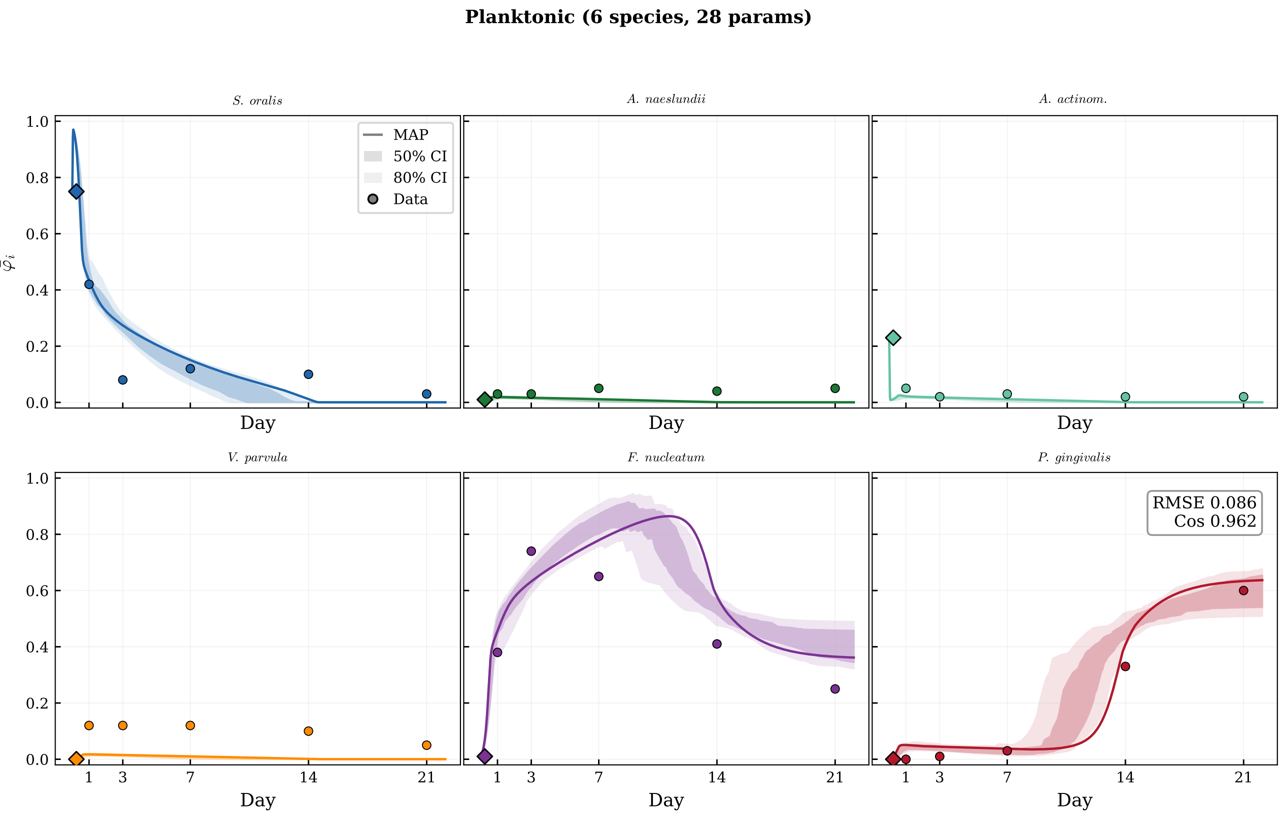

Figure 1: Planktonic posterior predictive fit (RMSE = 0.086). MAP trajectory (line), 50%/80% credible intervals (shaded), digitized data (circles), and initial condition (diamond at Day 0.25). The model captures the So→Fn→Pg succession pattern. *F. nucleatum* peaks at Day 3 (∼74% observed, ∼55% MAP) before declining as *P. gingivalis* rises via the Hill-gated mechanism ($K_{\text{hill}} = 0.115$). The main residual is the Fn peak underestimation; the CI bands cover most data points.

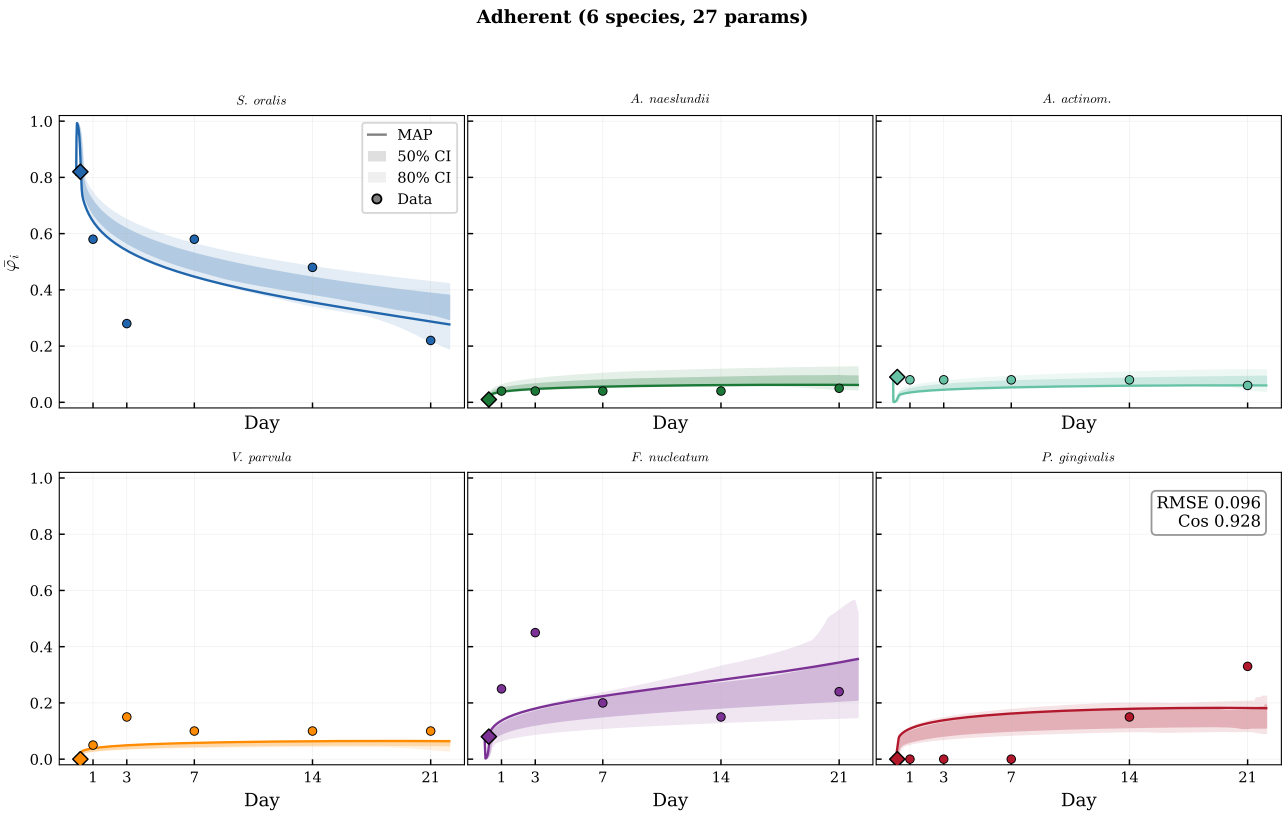

Figure 2: Adherent posterior predictive fit (RMSE = 0.096). The adherent biofilm shows slower dynamics than planktonic. *S. oralis* dominates longer (Day 7 recovery visible in data), and *P. gingivalis* emergence is delayed relative to planktonic. The model reproduces the monotonic So decline and gradual Pg increase well. *F. nucleatum* and *V. parvula* remain at low levels ($< 15\%$), consistent with surface-attached biofilm being spatially structured.

4 Interaction Matrix A (6×6)

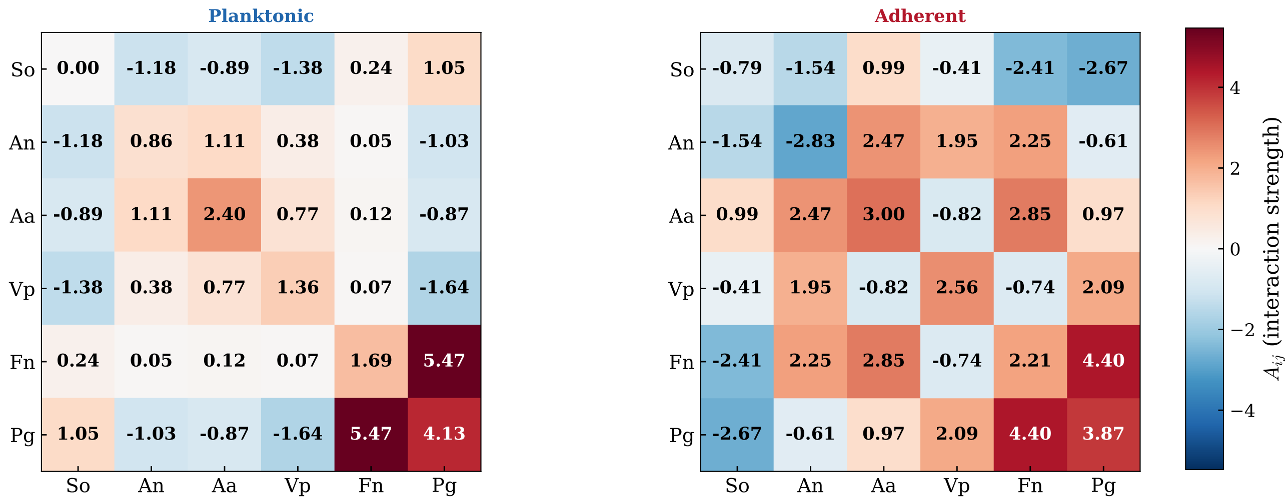

Figure 3: MAP interaction matrices. Red = mutualistic ($A_{ij} > 0$), blue = competitive ($A_{ij} < 0$). Both conditions share strong Fn–Pg mutualism ($a_{56} \approx 4$–6), reflecting the well-documented physical coaggregation between these species. Planktonic shows stronger So–An mutualism ($a_{12} = 3.65$) and So–Vp competition ($a_{14} = -3.70$), while adherent exhibits So–An competition ($a_{12} = -1.54$) — suggesting surface attachment fundamentally alters early-colonizer interactions. *A. actinomycetemcomitans* (Aa) shows condition-dependent behavior: weakly interactive in planktonic but moderately mutualistic with An/Vp in adherent.

5 Interaction Parameter Posteriors (a_{ij})

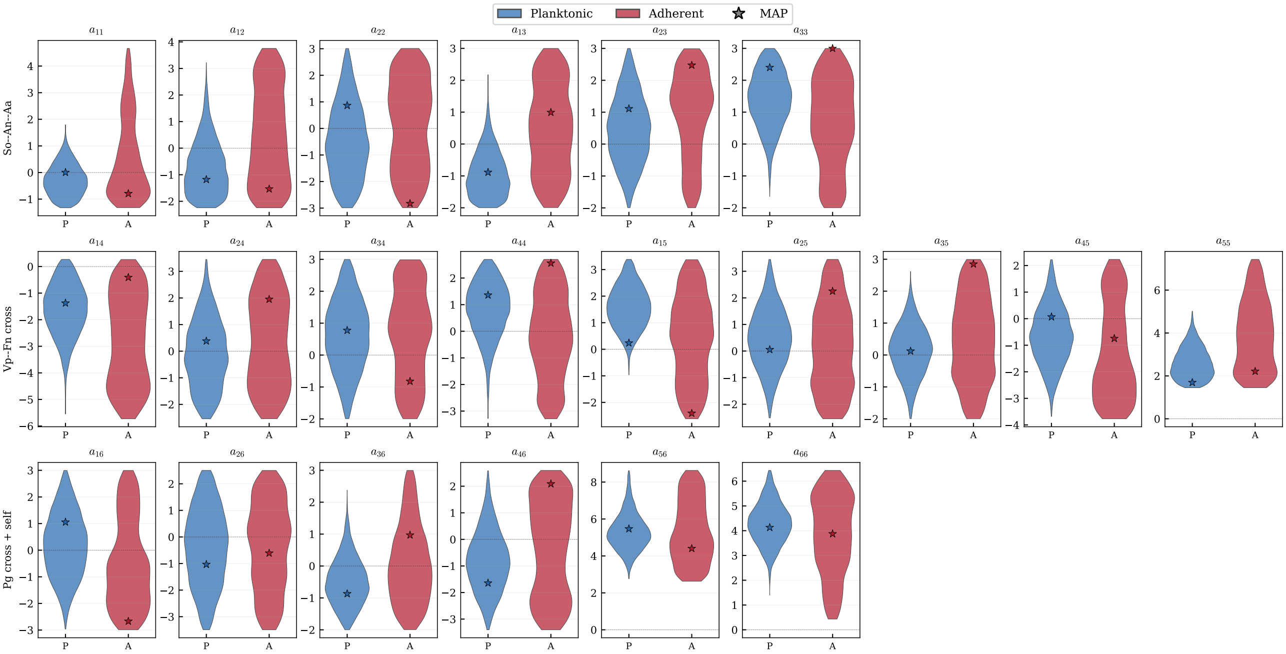

Figure 4: Posterior distributions of all 21 interaction parameters a_{ij}. P = Planktonic, A = Adherent. Star = MAP estimate. Top row: So–An–Aa block (early colonizer self- and cross-interactions). Middle: Vp–Fn cross-interactions (secondary colonizer coupling with all species). Bottom: Pg interactions (late colonizer). The Fn–Pg coupling (a_{56}) is consistently positive and tightly constrained in both conditions, supporting the Hill-gated bridging hypothesis. Several parameters show bimodal or wide posteriors (e.g., a_{14}, a_{25}), indicating partial non-identifiability in these interaction channels.

6 Growth Rates μ_i

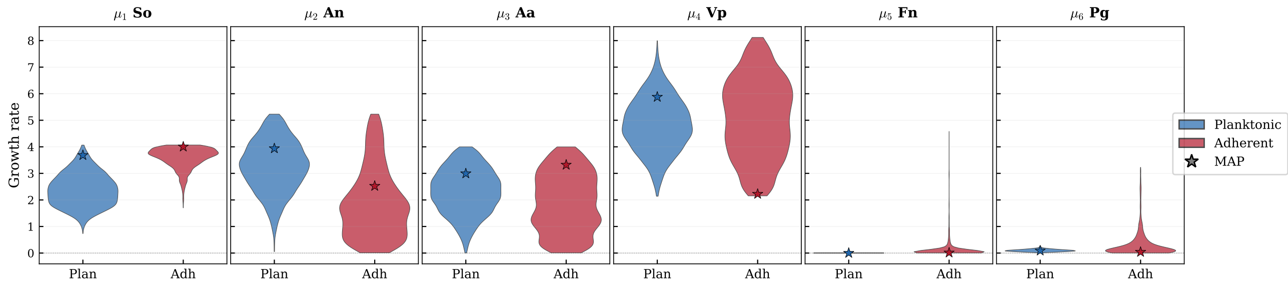

Figure 5: Posterior growth rate distributions (μ_i). Star = MAP. Both conditions assign high μ to *S. oralis* and *V. parvula*, consistent with their role as fast-growing early/secondary colonizers. *F. nucleatum* and *P. gingivalis* have lower μ values, relying on interaction terms (A_{ij}) rather than intrinsic growth to achieve dominance — particularly *P. gingivalis* which requires the Fn-mediated Hill gate to proliferate. *A. actinomycetemcomitans* shows near-zero μ in planktonic but positive in adherent, reflecting its surface-adhesion advantage.

7 Observed vs Predicted

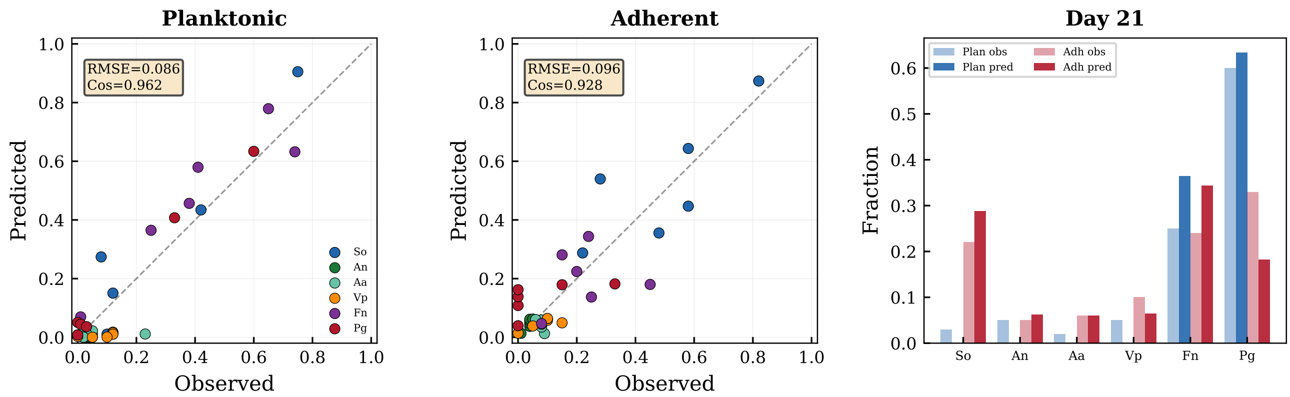

Figure 6: Left/center: observed vs predicted scatter for planktonic and adherent. Points near the diagonal indicate good fit; deviation above/below indicates over-/under-prediction. Adherent shows tighter clustering around the diagonal. Right: Day 21 equilibrium composition. Planktonic is Pg-dominated (∼60%), while adherent reaches a more balanced three-way split (So/Fn/Pg each ∼25–33%). The model captures this qualitative difference between the two environments.

8 Residual Analysis

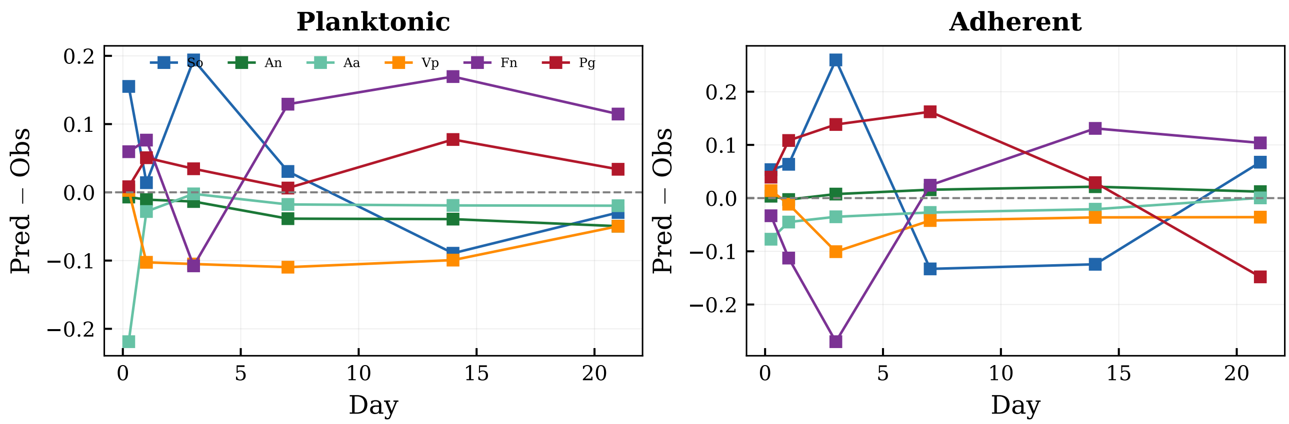

Figure 7: Per-species residuals (pred − obs) over time. Planktonic: largest residual is Fn at Day 3 (−0.19, underestimation of the Fn peak) and Pg at Day 14 (−0.18, delayed Pg prediction). Adherent: residuals are smaller and more uniform, with the largest at Day 3 for So (+0.11, overestimation due to the Day 7 reversal that the ODE cannot fully capture). All residuals are within ±0.2 for both conditions.

9 MAP Parameters (27)

Table 6: MAP θ

Idx	TeX	Name	Planktonic	Adherent
0	a_{11}	a(So-So)	+0.003	-0.791
1	a_{12}	a(So-An)	-1.179	-1.535
2	a_{22}	a(An-An)	+0.865	-2.828
3	a_{13}	a(So-Aa)	-0.888	+0.991
4	a_{23}	a(An-Aa)	+1.108	+2.474
5	a_{33}	a(Aa-Aa)	+2.400	+2.998
6	a_{14}	a(So-Vp)	-1.376	-0.410
7	a_{24}	a(An-Vp)	+0.382	+1.949
8	a_{34}	a(Aa-Vp)	+0.774	-0.823
9	a_{44}	a(Vp-Vp)	+1.359	+2.559
10	a_{15}	a(So-Fn)	+0.235	-2.409
11	a_{25}	a(An-Fn)	+0.053	+2.251
12	a_{35}	a(Aa-Fn)	+0.122	+2.848
13	a_{45}	a(Vp-Fn)	+0.067	-0.744
14	a_{55}	a(Fn-Fn)	+1.692	+2.213
15	a_{16}	a(So-Pg)	+1.053	-2.669
16	a_{26}	a(An-Pg)	-1.033	-0.609
17	a_{36}	a(Aa-Pg)	-0.869	+0.969
18	a_{46}	a(Vp-Pg)	-1.644	+2.089
19	a_{56}	a(Fn-Pg)	+5.468	+4.404
20	a_{66}	a(Pg-Pg)	+4.129	+3.874
21	μ_1	(So)	+3.683	+4.003
22	μ_2	(An)	+3.939	+2.523
23	μ_3	(Aa)	+2.993	+3.321
24	μ_4	(Vp)	+5.877	+2.222
25	μ_5	(Fn)	+0.000	+0.013
26	μ_6	(Pg)	+0.087	+0.037
27	K_{hill}	K$_hill$	+0.115	—